

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4.	(a)	(i)		Any 2 × 1 from: distances between burning food and bottom of tube – (otherwise variable heat loss) (1) masses of food used – (different masses release different amounts of energy) (1) volumes/mass of water in tube – (different volumes give different temperature rise) (1) surface area of food – (different amount of heat produced/more complete burning) (1) type of food– (different foods release different amounts of energy/ contain different amounts of energy) (1)	2			2		2
		(ii)		Water in tube does not absorb all the heat /heat lost to surroundings (1) Incomplete combustion of food (1)			2	2		2
		(iii)		Any 3 × (1) from: Provides oxygen to encourage complete combustion (1) Jacket of air surrounds the combustion chamber - insulating (1) Copper coil for better conduction/heat transfer to water (1) Stirrer to mix warm and cold water (1)		3		3		3
	(b)			$20 \times 15.4 \times 4\,200$ (1) $\text{Ans} / 1\,000 = 1\,293.6 \text{ J}$ (1) $1\,293.6 \text{ (ecf)} \div 12.4 = 104.3 \text{ (J/g)}$ (1) $1\,293.6$ on ans line (2) $1\,293\,600$ on ans line (1) $104\,300$ on ans line (2)		3		3	3	3
				Question 4 total	2	6	2	10	3	10